2021 ESC Heart Failure — RAG Knowledge Source

English • Text only • No tables • Retrieval-oriented condensed source

Source identity. McDonagh TA, Metra M, Adamo M, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. European Society of Cardiology / European Heart Journal, 2021.

Purpose. This document is a clean, condensed representation of clinically useful facts from the ESC guideline. It is designed for retrieval and question answering. References, long evidence discussions, repeated prose, and unrelated material are excluded.

RAG note. Each section below contains short, self-contained facts. Numerical thresholds remain attached to their clinical context.

1. Heart Failure Definition

Heart failure (HF) is a clinical syndrome associated with symptoms and signs caused by a structural and/or functional cardiac abnormality.

Symptoms and signs alone are not sufficient to establish the diagnosis. Objective evidence of cardiac structural or functional abnormality is required.

Common symptoms include breathlessness, fatigue, and ankle swelling.

Possible signs include elevated jugular venous pressure, pulmonary crackles, and peripheral edema.

2. Heart Failure Phenotypes

HFrEF: heart failure with reduced ejection fraction. LVEF is ≤40%.

HFmrEF: heart failure with mildly reduced ejection fraction. LVEF is 41–49%. Symptoms and/or signs of HF are present; raised natriuretic peptides and/or relevant structural abnormalities support the diagnosis.

HFpEF: heart failure with preserved ejection fraction. LVEF is ≥50%, with symptoms and signs plus objective evidence of relevant structural or functional abnormalities and/or raised natriuretic peptides.

LVEF means left ventricular ejection fraction and is commonly assessed by echocardiography.

3. Chronic Heart Failure Diagnosis

When HF is suspected, assess symptoms and/or signs, risk factors, previous cardiovascular disease, and the ECG.

An abnormal ECG increases the likelihood of HF. A normal ECG makes HF less likely.

In the non-acute diagnostic pathway, NT-proBNP ≥125 pg/mL or BNP ≥35 pg/mL prompts further evaluation.

Echocardiography is used to assess cardiac structure and function and to measure LVEF.

After objective abnormalities are identified, define the HF phenotype as HFrEF, HFmrEF, or HFpEF.

The underlying etiology should then be investigated because it can influence management.

4. NYHA Functional Classification

NYHA Class I: no limitation of physical activity. Ordinary physical activity does not cause undue symptoms.

NYHA Class II: slight limitation of physical activity. The patient is comfortable at rest, but ordinary physical activity causes symptoms.

NYHA Class III: marked limitation of physical activity. The patient is comfortable at rest, but activity less than ordinary causes symptoms.

NYHA Class IV: unable to carry out physical activity without discomfort. Symptoms may be present at rest and increase with activity.

5. Major Heart Failure Etiologies

Important causes and etiologies to investigate include:

- Coronary artery disease and myocardial infarction.
- Hypertension.
- Valvular heart disease.
- Arrhythmias.
- Dilated, hypertrophic, restrictive, and arrhythmogenic cardiomyopathies.
- Congenital heart disease.
- Myocarditis and selected infectious causes.
- Peripartum cardiomyopathy.
- Takotsubo syndrome.
- Relevant toxins or cardiotoxic exposures.

6. Diagnostic Tests

ECG: identifies electrical and structural clues. A normal ECG makes HF less likely.

BNP / NT-proBNP: natriuretic peptides used in the HF diagnostic pathway.

Echocardiography: assesses cardiac structure and function and measures LVEF.

Coronary assessment: coronary angiography, CT coronary angiography, or stress imaging may be used when CAD is suspected.

Ambulatory ECG: may be used when an arrhythmia is suspected.

Cardiac magnetic resonance: can characterize cardiomyopathies, myocarditis, and other structural abnormalities.

Laboratory tests: selected tests help identify contributing conditions and complications.

7. Acute Heart Failure

Acute heart failure (AHF) is a separate diagnostic and management pathway from chronic/non-acute HF.

For suspected AHF, rule-out thresholds are BNP <100 pg/mL, NT-proBNP <300 pg/mL, or MR-proANP <120 pg/mL.

For NT-proBNP, age-specific rule-in values include >450 pg/mL for age <55 years, >900 pg/mL for age 55–75 years, and >1800 pg/mL for age >75 years.

Initial laboratory assessment may include troponin, serum creatinine, electrolytes, blood urea nitrogen or urea, TSH, and liver function tests.

D-dimer is considered when acute pulmonary embolism is suspected. Procalcitonin may be considered when infection or pneumonia is suspected. Arterial blood gas analysis is used when respiratory distress requires precise oxygen and carbon dioxide assessment. Lactate and pH are relevant when cardiogenic shock or hypoperfusion is suspected.

Comprehensive echocardiography is part of the evaluation after AHF is confirmed or strongly suspected.

8. Acute Pulmonary Edema

Acute pulmonary edema is related to lung congestion.

Clinical features can include dyspnea with orthopnea, respiratory failure with hypoxaemia or hypercapnia, tachypnea above 25 breaths/min, and increased work of breathing.

When indicated, treatment may include oxygen or non-invasive respiratory support, intravenous diuretics, and intravenous vasodilators when systolic blood pressure is sufficiently high.

If acute pulmonary edema is associated with low cardiac output in advanced HF, inotropes, vasopressors, and/or mechanical circulatory support may be required.

9. Treatment — Core Retrieval Facts

HFrEF: major guideline-directed drug classes include ACE inhibitors or ARNI, beta-blockers, mineralocorticoid receptor antagonists, and SGLT2 inhibitors. Diuretics are used to relieve congestion when fluid retention is present.

HFmrEF: diuretics are recommended when congestion is present. Other treatments may be considered according to the clinical context.

HFpEF: management focuses on identifying and treating underlying causes, cardiovascular risk factors, and comorbidities, with treatment of congestion when present.

Treatment choices depend on the individual clinical situation, contraindications, renal function, blood pressure, and comorbidities. Detailed dosing is intentionally omitted from this RAG source.

10. Advanced Heart Failure

Advanced HF requires all specified criteria to be present despite optimal medical treatment.

Severe and persistent symptoms are defined as NYHA class III or IV.

Severe cardiac dysfunction may include LVEF ≤30%, isolated right ventricular failure, non-operable severe valve abnormalities, non-operable severe congenital abnormalities, or persistently high/increasing BNP or NT-proBNP together with severe LV diastolic dysfunction or structural abnormalities consistent with HFpEF.

Severe recurrent events include pulmonary or systemic congestion requiring high-dose intravenous diuretics, low-output episodes requiring inotropes or vasoactive drugs, or malignant arrhythmias causing more than one unplanned visit or hospitalization in the previous 12 months.

Severely impaired exercise capacity includes inability to exercise, a 6-minute walk test distance <300 m, peak VO_2 <12 mL/kg/min, or peak VO_2 <50% of predicted when the limitation is considered cardiac in origin.

11. RAG Retrieval Rules

Use one main clinical idea per chunk.

Keep numerical thresholds together with the condition they describe, especially when acute and non-acute thresholds differ.

Keep abbreviations explicit: HF, HFrEF, HFmrEF, HFpEF, LVEF, BNP, NT-proBNP, NYHA, AHF.

Separate diagnosis, phenotype classification, etiology, treatment, acute HF, and advanced HF into different chunks.

Do not ingest reference lists, author lists, page headers, URLs, or repeated guideline boilerplate as clinical knowledge.

Recommended metadata: source_id=ESC_HF_2021; topic; section; keywords; original page if available.

12. Example Retrieval Unit

Topic: Chronic HF diagnosis.

Question: What is the basic diagnostic principle for chronic heart failure?

Answer: Diagnosis requires compatible symptoms and/or signs together with objective evidence of cardiac structural or functional abnormality.

Source ID: ESC_HF_2021.

Keywords: heart failure, diagnosis, symptoms, signs, cardiac dysfunction, LVEF, echocardiography.